

TrES-5b

R-1628

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220824 · created 2026-08-02T05:25:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459815.8744 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.052
Transit depth (Rp/Rs)^2	1.17 +/- 1.12 [%]
Orbital Inclination (inc)	81.26 +/- 2.93
Airmass coefficient 1 (a1)	1.04 +/- 0.86
Airmass coefficient 2 (a2)	0.14 +/- 0.16
Scatter in the residuals of the lightcurve fit is	96.32 %
Best Comparison Star	#2 - [569, 263]
Optimal Aperture	1.07
Optimal Annulus	4.27
Transit Duration (day)	0.039 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.052 vs 0.142 (deviation 0.28 σ)
Duration fit vs published	0.039 d vs 0.0758 d (deviation 0.55 σ)
Mid-time O-C	-9.1 min
Depth SNR	0.9
Residual scatter	96.32 %

Light curve

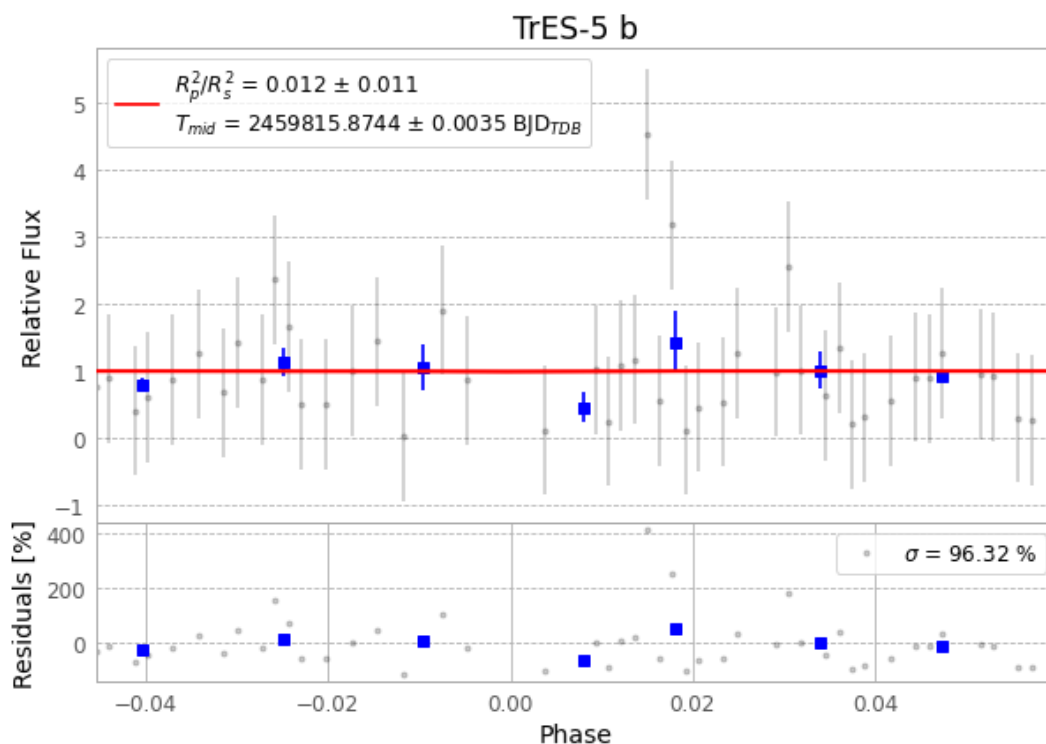


Figure 1 — FinalLightCurve_TrES-5 b_2022-08-24.png (SHA-256 26f4096262dfff1a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9319e4258d9851a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), TRES-5220824071813.FITS → TRES-5220824110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220824.log (3b3c4038c959...), FinalLightCurve_TrES-5 b_2022-08-24.png (26f4096262df...), FinalLightCurve_TrES-5 b_2022-08-24.csv (c6a65be554e7...)

Compute resources

Wall time 245.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:25Z.